AUTUMN END SEMESTER EXAMINATION-2014

5th Semester MCA

SOFT COMPUTING
MCA - 522

Regular - 2012 Admitted batch & Back

Full Marks: 60 **Time: 3 Hours**

Answer any six questions including question No. 1 which is compulsory.
The Figures in the right hand margin indicate full marks.
Candidates are required to give their own words as far as practicable and all parts of a question should be answered at one place only.

1. a) Define the role of membership function in fuzzy set. [2x10]

   b) What is fuzzy proposition?

   c) Define fuzzification and difuzzification.

   d) How the operations on fuzzy relations are different from crisp relations?

   e) Discuss about the genetic algorithm operators and parameters.

   f) Discuss about the fitness function in Genetic Algorithm?

   g) What do you understand by Activation function in Neural Network?

   h) What is Recurrent Network?

   i) How hybrid systems are classified in soft computing?

   j) Discuss basic Artificial Neural Network Models.

2. Let U be the universe of military aircraft of interest' defined below: [8]

U = {a10, b52, c130, f2, f9}

Let A be the fuzzy set of the bomber class aircraft:

$$A= \frac{0.3}{a10}+\frac{0.4}{b52}+\frac{0.2}{c130}+\frac{0.1}{f2}+\frac{1}{f9}$$

Let A be the fuzzy set of the fighter class aircraft:

$$BA= \frac{0.1}{a10}+\frac{0.2}{b52}+\frac{0.8}{c130}+\frac{0.7}{f2}+\frac{0}{f9}$$

Find the fuzzy operations A∩B, $\overline{A}$, $\overline{A} \cup \overline{B}, \overline{A \cap B}$

3. a) Let R, S be defined on the sets {1, 3, 5} × {1, 3, 5} and R :{(x, y)| y=x+2}, S :{(x, y) |x<y} find the relational matrices and the Max-min composition R ∘ S . [4x2]

b) Let P= {a, b, c, d} Q= {p, q, r, s} and

A= {(a, 0), (b, 0.7), (c, 0.5), (d, 1)}
B= {(p, 0.3), (q, 1), (r, 0.6), (s, 0)}
C= {(p, 0.0), (q, 0.5), (r, 1), (s, 0.3)}

Determine the fuzzy implication relations

i. if x in A then y is in B and
ii. if x is A then y is B else y is C

4. a) Discuss different encoding methods in Genetic Algorithm? [4x2]

b) Differentiate between supervised and un-supervised learning in neural network?

5. a) How offspring are produced and outline the basic of Genetic Algorithm? [4x2]

   b) What do you understand by "Tournament Selection" with reference to Genetic Algorithm? How does it overcome the demerit of "Roulette Wheel Selection"?

6. a) Differentiate between single layer and multilayer feed forward network? [4x2]

   b) Define the basic model and characteristics of Neural Network?

7. a) $X=\{x_1, x2, x3\}$ $Y=\{y_1, y2\}$ $Z=\{z_1, z2, z3\}$ [4x2]

   Let R be a fuzzy relation

	y_1	y_2
x_1	0.7	0.2
x_2	0.2	0.9
x_3	0.7	0.6

   and S be a fuzzy relation

	Z_1	Z_2	Z_3
Y_1	0.9	0.4	0.7
Y_2	0.5	0.2	0.4

   What is max-min composition of $R \circ S$?

   b) How sequential hybrid system is different from Auxiliary and Embedded Hybrid System?

8. Short Notes (answer any two) [4x2]
   a) Elitism
   b) Centroid Method
   c) Learning Rules in Neural Network

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